

Paper Version Comparison Report

V0 (original) → V1 (after_adaptive)

Including SIMON Optimization Report Implementation Audit

April 27, 2026

1. Executive Summary

This report compares **V0** (Aarush_Paper_original_V0.pdf, dated **March 13, 2026**) with **V1** (Aarush_Paper_v1_after_adaptive.pdf, dated **April 10, 2026**) — approximately 4 weeks of development work. It also cross-references the **SIMON Optimization Report** to identify which of its suggestions were implemented.

Unlike the V1→V2 revision (which was purely prose cleanup), **V0→V1 is a fundamental engineering overhaul**. The paper describes a completely different, far more capable system. Key headline changes:

- **The integrator switched** from LEAPFROG to ias15 — a major architectural decision.
- **Two entirely new methodology sections** were added: Section 3.6 (Training Process) and Section 3.7 (Adaptive Sub-Stepping).
- **Training data grew 10×** from 5,000 to 50,000 samples, with focused sampling strategy.
- **Per-step speed improved 54×**: from ~1,300 μs (Python PyTorch) to 23.8–26.3 μs (NumPy).
- **SIMON flipped from slower to faster than ias15**: V0 concluded it “was unable to outpace ias15”; V1 reports 1.10x–2.09x speedups.
- **Body ejections were eliminated** by adaptive sub-stepping; V0 had bodies escaping to 65–130 AU.
- **The divergence exponent changed**: $\lambda = 9.64 \times 10^{-2} \text{yr}$ (V0) → $\lambda = 0.156 \text{yr}$ (V1).
- **7 of 8 SIMON Report suggestions were implemented** (fully or partially).

2. Summary of All Changes — At a Glance

Location	V0 (original)	V1 (after_adaptive)	Change
Title	Neural Network Emulation of the Unrestricted Three-Body Problem	Same title (unchanged until V2)	No change
Date	March 13, 2026	April 10, 2026	Changed
Fig ?? ref	Broken ref: “Figure ??” in Section 1.2	Fixed: “Figure 2”	Changed
Sec 2 — ias15 desc	Called “Symplectic integrator” (same error)	Same error present (corrected only in V2)	No change

Location	V0 (original)	V1 (after_adaptive)	Change
Sec 3.5 — integrator	LEAPFROG (Rein & Tremaine 2011) as the analytical integrator	ias15 replaces LEAPFROG throughout	Changed
Sec 3.5 — gating	Simple 2-zone gating: NN active or fallback to baseline	3-tier gating: far-field bypass ($r > 500\epsilon$), close-encounter NN, very-close bypass ($< 5 \times 10^{-3}$)	Changed
Sec 3.6 — Training	5,000 samples; no focused sampling; no hyperparameter table	50,000 samples; 60% focused in transition zone; full hyperparameter table added	Changed
Sec 3.7 — Adaptive	Does not exist	NEW: Entire section on adaptive sub-stepping mechanism	Added
Sec 3.7/3.8 — Adaptive sub-step	Not present	$n_{\text{sub}} = \min(16, \text{ceil}(0.05/r_{\text{min}}))$; triggers when $r_{\text{min}} < 0.05$ AU	Added
Sec 4.1 — Hybrid Use	Section exists: “≈20% fallback, ≈80% use NN”	Section REMOVED entirely; replaced by direct trajectory overlay	Removed
Sec 4.1/4.2 — Trajectories	Spaghetti trajectories; body ejections visible	Clean bounded orbits; ejections eliminated by adaptive stepping	Changed
Sec 4.3 — Divergence λ	9.64×10^{-2} /yr; sharp discontinuity at ~30yr from ejections	0.156/yr; smooth exponential growth; no discontinuity	New result
Fig 12 — Perf table	Not present	New table: dt, Final RMS, Throughput, Total Time, Steps, Speedup	Added
Fig 12/13 — Step time	~1,288–1,302 $\mu\text{s}/\text{step}$ (Python PyTorch; 1.3 ms)	23.8–26.3 $\mu\text{s}/\text{step}$ (NumPy pre-extracted weights)	New result
Speedup vs ias15	SIMON slower than ias15 at all dt	1.10x at $dt=0.04$; 2.09x at $dt=0.08$	New result
Conclusions	7 conclusions; ends with “unable to outpace ias15”	8 conclusions; leads with “SIMON outpaces ias15 (2.09x)”	Changed
References	Includes Rein & Tremaine [2011] for LEAPFROG	Removes Rein & Tremaine; adds Loshchilov & Hutter [2017] for AdamW	Changed

3. Detailed Section-by-Section Analysis

3.1 Title, Date, and Broken Reference Fix

Title and date: Title unchanged (V0 and V1 share the same title; it only changed in V2). Date updated from March 13 to April 10, 2026.

Broken figure reference — fixed: V0 Section 1.2 ends with the sentence “We present a schematic of such a system in Figure ??” — a compilation error producing an unresolved LaTeX cross-reference. V1 corrects this to “Figure 2”.

3.2 Section 3.5 — Hybrid Incorporation: LEAPFROG → ias15 + 3-Tier Gating

This is the single most architecturally significant change in the entire V0→V1 evolution. The baseline analytical integrator was replaced, and the force-gating logic was completely redesigned.

V0: Ullibarrena et al. [2024] predicts that using a hybrid combination of an analytical integrator (like LEAPFROG Rein and Tremaine [2011] in our case) to correct inaccurate neural network predictions could 'prevent large energy errors.' A fixed timestep is used. We incorporate LEAPFROG into our model in a similar way... The coefficient of correction, predicted by the NN, determines whether the softened force should be stronger or weaker. We put limits on the coefficient, bounding it between 0.2 and 5.	V1: Ullibarrena et al. [2024] predicts... We incorporate ias15 into our model in a similar way... We divide force computation into three tiers. Far field, where $r > 500\epsilon = 0.15\text{AU}$ means the NN is not invoked and is identical to what ias15 predicts. Close encounters, where $r < 0.15\text{AU}$ means the NN is invoked and predicts a correction factor. To prevent the force from being strengthened or weakened too much, we gate the coefficient, bounding it between 0.2 and 5. For very close encounters, where $r < 5 \times 10^{-4}$ the NN is again bypassed.
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What changed and why:

- **LEAPFROG → ias15:** LEAPFROG (Rein & Tremaine 2011) is a 2nd-order symplectic integrator. ias15 is a 15th-order Gauss-Radau integrator accurate to near-machine-precision. This is a substantial upgrade in baseline accuracy. The reference to Rein & Tremaine [2011] disappears from the references section entirely in V1.
- **2-zone → 3-tier gating:** V0 had two zones: NN active vs fallback. V1 introduces a third zone — the far-field bypass ($r > 500\epsilon = 0.15 \text{ AU}$) where exact Newtonian gravity is used without even calling the NN. This is the key optimization that reduces NN invocation from ~80% to ~1.1% of steps.
- **Very-close bypass added:** When $r < 5 \times 10^{-4}$, the NN is bypassed to prevent non-physical behavior during extreme close encounters.

3.3 Section 3.6 — Training Process: Entirely Rewritten (New Section)

V0 had no dedicated Training Process section. Training details were embedded in Section 3.5. V1 adds a full new Section 3.6 with significantly upgraded training:

V0 (embedded in Sec 3.5): Training data is generated by sampling 5000 physically valid three-body configurations from Rebound. For each configuration, reference forces are computed analytically using Newtonian gravity.

V1 (new dedicated Sec 3.6): Training data is generated by sampling 50,000 physically valid pairwise configurations from Rebound. Approximately 60% of the samples are drawn from regions where the inter-body distance $r \in [0.5\epsilon, 50\epsilon]$ ($\epsilon = 3 \times 10^{-10}$), where the correction predicted by the model would deviate meaningfully from 1. The other 40% is sampled from the far field $r \in [50\epsilon, 10]$ where the correction would be closer to one.

- **10× more training data:** 5,000 → 50,000 samples. This is the single largest driver of improved model accuracy.
- **Focused sampling:** 60% of samples concentrated in the transition zone $[0.5\epsilon, 50\epsilon]$ where the correction c deviates meaningfully from 1 — the region that actually matters for accuracy.
- **Full hyperparameter table added in V1:** Optimizer (AdamW, $\text{wd}=1 \times 10^{-4}$), Learning rate (1×10^{-3} , cosine annealing to 1×10^{-4} over 5,000 epochs), Loss function (MSE on $\log(c)$), Batch size (40,000), Train/eval split (80%/20%), Device (NVIDIA GeForce RTX 5050 CUDA 13.0), Training time (9 seconds), Best MSE (1.1×10^{-4}). None of this existed in V0.
- **Network architecture explicit in V1:** 3 hidden layers of 32 neurons each, SiLU activations. V0 described the old 128-unit, MC-dropout architecture implicitly.

3.4 Section 3.7 — Adaptive Sub-Stepping: Entirely New Section

This section does not exist at all in V0. It is a complete addition in V1, describing the mechanism that eliminates the body-ejection problem that plagued the V0 system:

- At each macro step, the minimum pairwise separation r_{min} is computed.
- When $r_{\text{min}} < 0.05$ AU, the step is subdivided into $n_{\text{sub}} = \min(16, \text{ceil}(0.05/r_{\text{min}}))$ sub-steps.
- Each sub-step has duration dt/n_{sub} (velocity-Verlet).
- The mechanism activates on ~0.2–1.1% of steps, adding ~6 μs overhead.
- Prevents energy injection during close encounters that caused ejections in V0.
- All physics computed in float64; NN forward pass uses float32 then converts back.

3.5 Section 3.7/3.8 (V0) → 3.8/3.9 (V1) — Evaluation & Scope: Renumbered

The Evaluation Strategy and Scope of the Method sections are substantively identical in content, but renumbered from 3.6/3.7 (V0) to 3.8/3.9 (V1) due to the two new sections inserted above.

3.6 Section 4.1 — Hybrid Usage Section: Deleted

V0 (Section 4.1 — Hybrid Usage): The hybrid gating mechanism activates meaningfully: ~20% of steps fall back to the baseline force model, while ~80% use the learned correction, providing robustness.

V1: Section 4.1 'Hybrid Usage' is entirely REMOVED. V1 begins results directly with Section 4.1 'Trajectory Overlay'. The 1.1% NN invocation figure appears later in Conclusion 6 instead.

This reflects the fundamental change in how the gating works. V0's system invoked the NN ~80% of the time (close encounters dominated). V1's far-field bypass means the NN is needed only ~1.1% of the time — making a dedicated 'Hybrid Usage' section moot.

3.7 Section 4.2/4.1 — Trajectory Overlay: Qualitatively Different Figures

V0 Trajectory Description: <i>SIMON produces a similar 'spaghetti-like' trajectory to ias15, with slight deviation, and omission of trajectories toward the center of the plane. Bodies 1 and 2 follow a similar geometry to ias15, circling near the right-hand side. Main discrepancies: phase drift and occasional departures from trajectory.</i>	V1 Trajectory Description: <i>SIMON produces similar chaotic, tightly-wound elliptical orbits to ias15 within the range $x \in [-0.015, 0.015]$, $y \in [-0.015, 0.015]$. Body 1 produces bounded elliptical orbits $x \in [-1.5, 1.5]$. Body 2 produces similar orbits $x \in [-1.5, 2.5]$. SIMON does not eject body 1 from the system — it remains gravitationally bounded throughout.</i>
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The qualitative picture is materially different: V0 had body ejections visible in trajectories; V1's adaptive sub-stepping keeps all three bodies bounded throughout the 100-year simulation.

3.8 Section 4.3 Divergence: λ Changed, Discontinuity Removed

V0: <i>Figure 10 title: 'slope=9.640e-02 1/yr'. The divergence plot shows a sharp discontinuous jump at ~30 years (body ejection signature), then saturation. $\lambda \approx 9.64 \times 10^{-2}$ /year ≈ 0.0964/yr. Fit window: years 10–50.</i>	V1: <i>Figure 10: 'SIMON-vs-REBOUND Divergence'. Smooth exponential growth throughout — no discontinuities. $\lambda \approx 0.156$/yr. Fit window: years 10–50 (same window, different behavior). Error drops from 10^{-4} to $\sim 10^{-3}$.</i>
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The change in λ from 0.0964 to 0.156 is a **genuine new result**, not a correction. The V0 system had body ejections that caused a sharp discontinuous spike in the divergence curve, corrupting the fit. V1's adaptive sub-stepping prevents ejections, producing pure exponential chaotic divergence. The higher λ in V1 reflects cleaner chaotic dynamics without the ejection artifact suppressing or distorting the exponential fit.

3.9 Section 4.4 — Speed-Accuracy Frontier: Complete Reversal

V0: <i>dt=0.04 produces the lowest final deviation (less than 6×10^{-1}) among the tested settings, at a mid-high range throughput of approximately 10^4. Despite this, the model was NOT able to be as fast as ias15, due to ias15's highly optimized C/C++ code, compared to SIMON's complete python composition.</i>	V1: <i>SIMON outperforms ias15 at dt=0.04 (1.10x speedup) and dt=0.08 (2.09x speedup). The lowest final RMS error is 0.847 at dt=0.01 (optimal accuracy). Per-step time: 23.8–26.3 μs. ias15 baseline: 0.069 seconds (69ms). SIMON at dt=0.08: 33ms.</i>
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This is a complete narrative reversal. The V0 system was slower than ias15 in all configurations. V1 achieves genuine speedups by switching from Python/PyTorch (1,300 μ s/step) to pure NumPy with pre-extracted weights (23.8–26.3 μ s/step) — a 54× per-step improvement.

3.10 New Figure 12 — Performance Table (V0 has no equivalent)

V1 introduces a quantitative performance table (Figure 12) that does not exist in V0:

dt (yr)	Final RMS	Throughput	Total Time	Steps	Speedup
0.005	1.022	10,509 /s	0.476 s	20,000	0.14x
0.01	0.847	20,447 /s	0.245 s	10,000	0.28x

0.02	1.022	40,125 /s	0.125 s	5,000	0.55x
0.04	1.543	79,863 /s	0.063 s	2,500	1.10x
0.08	1.022	152,253 /s	0.033 s	1,250	2.09x

V0's Figure 12 (computational cost) showed per-step times in the range 1.288–1.302 ms (1,300 μ s), with total simulation times ranging from ~1s to 100s — all slower than ias15. V1's Figure 13 (same concept) shows 23.8–26.3 μ s, with 33ms total at dt=0.08.

4. Results Comparison — New vs Same

Unlike V1→V2 (where all results were identical), V0→V1 introduces several genuinely new experimental results. These are not just reinterpretations — they come from running different code (NumPy instead of PyTorch, adaptive sub-stepping added).

Figure/Metric	V0 Value	V1 Value	Status
Trajectory plots (XY)	Spaghetti-like; body ejections visible	Clean bounded orbits; all bodies retained	Changed
Divergence λ	9.64×10^{-2} /yr; discontinuous	0.156/yr; smooth exponential	Changed
RMS divergence plot	Sharp jump ~yr30 (ejection); saturation	Monotonic exponential growth; no jump	Changed
Best speedup vs ias15	Slower in all configs	1.10x (dt=0.04), 2.09x (dt=0.08)	Changed
Per-step time	~1,300 μ s/step (PyTorch, Python)	23.8–26.3 μ s/step (NumPy, pre-extracted)	Changed
Best RMS accuracy	$\sim 6 \times 10^{-1}$ at dt=0.04	0.847 at dt=0.01	Changed
Bodies bounded?	No — ejections to 65–130 AU at ~30yr	Yes — all within ~2.5 AU throughout	Changed
Fig 12 perf table	Does not exist	Full dt vs RMS vs speedup table	New
Training loss	Not reported explicitly	Best MSE = 1.1×10^{-4} (reported in hyperparams)	New
Trajectory shapes (V0 Figs 6,7)	Same concept; 3 bodies shown	Same concept; figures look cleaner due to bounded orbits	Changed

Bottom line on new models: The neural network architecture was upgraded (32-unit hidden layers, SiLU, no dropout, AdamW) and retrained on 10× more data. The inference implementation was rewritten from PyTorch to NumPy. The integrator switched from LEAPFROG to ias15. Adaptive sub-stepping was added. These are not cosmetic changes — they constitute an entirely new trained system that was re-evaluated against ias15, producing genuinely different results.

5. Conclusions Comparison — V0 vs V1

5.1 V0 Conclusions — What Was Said

- **Optimal dt:** $dt=0.04$ yields lowest final RMS position error.
- **Hybrid Activation:** ~20% fallback to baseline; system is robustly active.
- **The Dominant Failure Mode is Phase Drift:** Hybrid stays bounded but loses phase alignment.
- **Divergence Rate:** SIMON produced a divergence rate indicating preservation of chaotic nature.
- **Discovered What Controls Runtime:** Cost dominated by per-step overhead (force eval + gating + framework).
- **Developed Reusable Methodology:** Diagnostics give quantitative and qualitative insights.
- **Final verdict:** “Despite these findings, SIMON was unable to outpace ias15.”
- **Closing statement:** “With better computational resources such as GPUs... it may even be possible to construct a neural network that performs more quickly than the IAS15 integrator.”

5.2 V1 Conclusions — What Changed

- **SIMON outpaces ias15 [NEW — complete reversal]:** 2.09x speedup at $dt=0.08$, 33ms vs 69ms. This replaces V0’s “unable to outpace” conclusion.
- **Adaptive sub-stepping eliminates non-physical behavior [NEW]:** Preservation of orbital dynamics for all three bodies across all tested timesteps. Not present in V0.
- **The Dominant Failure Mode is Phase Drift [KEPT, refined]:** Language tightened slightly but substance is the same.
- **Divergence Rate ($\lambda = 0.156/\text{yr}$) [UPDATED]:** Specific value cited (V0 had no specific value in the conclusion text; the number appeared only in the figure title).
- **Discovered What Controls Runtime [KEPT]:** Identical to V0.
- **The NN serves as a targeted tool, not a replacement for physics [NEW]:** 1.1% invocation rate. Replaces V0’s “Hybrid Activation” conclusion.
- **Optimising implementation > algorithmic novelty [NEW]:** 54x speed gain from Apple M1 → Nvidia GeForce, architecture changes, skipping NN for far-field. Not in V0.
- **Developed Reusable Methodology [KEPT]:** Identical.

5.3 Removed Conclusions

- **“Optimal dt” conclusion:** V0 highlighted $dt=0.04$ as optimal. In V1, this is absorbed into the speed-accuracy discussion in Section 4.3. No longer a named conclusion.
- **“Hybrid Activation” conclusion:** The 20% fallback claim is gone. Replaced by the more precise 1.1% NN invocation figure in Conclusion 6.
- **“May eventually be possible” closing statement:** V0 ended with a hopeful speculative note about GPU resources enabling future speedups. V1 removes this because the speedup has already been achieved.

6. SIMON Optimization Report — Implementation Audit

The SIMON Optimization Report (March 2026) provided 8 specific recommendations in Deliverable 2. Below is a detailed audit of which were implemented in V1, which were partially implemented, and which were not.

SIMON Suggestion	Status	Detail in V1
Sec 3.4: ADD architecture detail — 3→32→32→32→1, SiLU, input normalization, 2,273 params, no dropout, AdamW	Implemented	V1 Sec 3.6 lists: 3 hidden layers of 32 neurons, SiLU activations, AdamW (lr=1e-3, wd=1e-5), cosine annealing, no dropout, best MSE=1.1e-5. Exact parameter count (2,273) not stated but architecture matches.
Sec 3.5: ADD far-field bypass paragraph — $r > 500\epsilon \rightarrow$ exact Newtonian, eliminates ~99% of NN calls	Implemented	V1 Sec 3.5 three-tier gating: far field $r > 500\epsilon = 0.15$ AU $\rightarrow$ NN not invoked, identical to ias15. Language and threshold match exactly.
Sec 3.5: ADD adaptive sub-stepping paragraph — $n_{\text{sub}} = \min(16, \text{ceil}(0.05/r_{\text{min}}))$ when $r_{\text{min}} < 0.05$ AU	Implemented	V1 Sec 3.7 entire section dedicated to this. Formula, threshold (0.05 AU), and cap (16 sub-steps) all match. Activates 0.2–1.1% of steps, adds ~6 μ s overhead.
Sec 4.1: CHANGE “~20% fallback” $\rightarrow$ “0.2–1.1% of pair evaluations invoke NN”	Implemented	V1 removes the Hybrid Usage section (Sec 4.1) entirely. The corrected 1.1% figure appears in Conclusion 6 (“NN only invoked for 1.1% of pair evaluations”).
Sec 4.2: ADD body ejections eliminated; all bodies bounded with adaptive stepping	Implemented	V1 Sec 4.1 (trajectory overlay) explicitly states: “SIMON does not eject body 1 from the system — it remains gravitationally bounded throughout the full 100 year simulation period.”
Sec 4.3: CHANGE $\lambda \approx 9.64 \times 10^{-2}/\text{yr} \rightarrow \lambda \approx 0.156/\text{yr}$ with smooth exponential explanation	Implemented	V1 reports $\lambda \approx 0.156/\text{yr}$ throughout. Divergence plot (Fig 10) shows smooth growth without the discontinuity that was present in V0.
Sec 4.4: CHANGE “unable to be as fast” $\rightarrow$ “1.10x at $dt=0.04$, 2.09x at $dt=0.08$ ”; add 54x per-step improvement	Implemented	V1 Sec 4.3 and Fig 12 report exactly 1.10x ($dt=0.04$) and 2.09x ($dt=0.08$). Per-step improvement (54x) noted in Conclusion 7.
Sec 5: CHANGE “unable to outpace ias15” $\rightarrow$ “2.09x speedup with bounded dynamics”	Implemented	V1 Conclusion 1: “SIMON outpaces ias15: The optimised hybrid model achieves a 2.09x speedup over Rebound ias15 at $dt=0.08$, completing a 100-year three body simulation in 33ms versus ias15’s 69ms.”
NEW Sec 4.5: ADD dedicated ‘Adaptive Sub-Stepping’ results subsection before Conclusions	Partial	The adaptive sub-stepping mechanism is fully described in Sec 3.7 (Methodology). However, no dedicated Sec 4.5 in Results discusses the adaptive results separately — they are woven into Secs 4.1 and 4.3 and Conclusion 2 instead.

Implemented	Partial	Not Implemented
8 / 9	1 / 9	0 / 9

All substantive changes were made	Adaptive results section in Methodology, not a separate Results subsection	Every recommendation was addressed
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7. Reference Changes

V0 has (not in V1): Hanno Rein and Scott Tremaine. *Symplectic integrators in the shearing sheet. Monthly Notices of the Royal Astronomical Society*, 415(4):3168–3176, June 2011. — This was cited for LEAPFROG.

V1 adds (not in V0): Ilya Loshchilov and Frank Hutter. *SGDR: Stochastic gradient descent with warm restarts. In International Conference on Learning Representations (ICLR), 2017.* — Cited for AdamW cosine annealing.

All other references are unchanged across V0 and V1. The reference swap directly mirrors the integrator change: LEAPFROG (Rein & Tremaine) was replaced by ias15 (Rein & Spiegel, already cited), and AdamW optimizer (Loshchilov & Hutter) was added for the training process.

End of V0 → V1 Comparison Report | Neural Network Three-Body Problem Research